

Mycroft — Chapter 2 Workbook

The Reallocation Principle: Completed Exercises & Your Own Mycroft Project

Prepared for: **Pranavi**

Project: **Your Own Mycroft** — Investment Research Desk

Thesis under audit: **Titan Company (India: TITAN)**

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Part 1 — LLM Exercises (1–3)

Exercise 1 — Label a real workflow's steps

Demonstrated using a monthly AR aging review as the working example:

Step	Label
Pull open invoices from the AR subledger	Preparation
Calculate days-outstanding per invoice	Preparation
Bucket into aging tiers (0-30, 31-60, 61-90, 90+)	Preparation
Flag accounts crossing a >90-day threshold	Preparation
Draft a summary paragraph on which customers drive the 90+ bucket	Transformation (not yet judgment)
Decide whether a flagged account needs a collections call vs. write-off	Judgment
Decide whether the write-off is material enough to disclose	Judgment
Controller signs off before report goes to CFO	Approval
Report is sent/filed	Release

Where models typically mislabel: the draft-summary step often gets mistaken for judgment because it reads like an analytical conclusion — it isn't; it's a transformation of already-flagged data into prose. Real judgment (call vs. write-off, materiality) happens after.

Exercise 2 — Prompt with explicit uncertainty markers

Prompt A (baseline): "Write a variance summary explaining why Q3 revenue declined 8%, based on the attached data."

Prompt B (with uncertainty markers): instructs the model to cite the specific row/field for any causal claim it can verify, and to write "[UNVERIFIED: possible driver — would need X to confirm]" for anything it cannot — explicitly stating that several UNVERIFIED tags are a better output than one clean but unsupported paragraph.

Expected difference: Prompt A produces a single fluent narrative — the fluency-trap failure from Chapter 1, confidently attributing the decline to a plausible-sounding cause. Prompt B forces separation of sourced fact from inference, functioning as an inline version of the Chapter 1 assessment artifact.

Exercise 3 — Design a phase gate

Applied to the AR aging workflow above:

- **Named approver:** Credit & Collections Manager (not "whoever's free").
- **What the recipe hands them:** aged AR table with flagged 90+ accounts, source export path/timestamp, draft summary with uncertainty markers.
- **Stop conditions:** no collections call initiated, nothing sent to customer, no write-off posted — recipe stops at the flagged, sourced table.
- **Defensible tomorrow:** the specific subledger export (date/timestamp), the aging-bucket logic, and a named account-owner confirmation for any write-off escalation.

Gap found on model review: the design didn't specify who owns changing the >90-day threshold itself for a customer on a known payment plan — a legitimate scope gap worth adding a rule for.

Part 2 — Your Own Mycroft Project

This chapter builds a research-effort plan on top of Chapter 1's claim-audit — deciding not just what's unresolved, but what's actually worth Pranavi's limited research time versus what to consciously ignore.

Exercise 1 — When to Use AI

- Inventorying a thesis into preparation vs. judgment steps — mechanical decomposition, correctable against the source text.
- Sketching which open claims could plausibly flip the decision — structured reasoning over supplied inputs, confirmed by the user.
- Drafting the time-cost side of the plan (roughly how long each piece of evidence would take to pull) — estimation, sanity-checked by experience.

Exercise 2 — When NOT to Use AI

- Deciding which claim actually moves the decision — depends on personal thesis weighting and risk tolerance, which the model cannot know.
- Deciding the research effort is "done" and the position is ready — no accountability for capital; cannot sense when evidence is genuinely thin.
- Judging whether an options signal is real positioning vs. noise — requires actual term-structure/microstructure context not given to the model.

Exercise 3 — Research Effort Plan (draft, then finalized by hand)

Original thesis audited (from Chapter 1): *Titan Company (India: TITAN)* — see Chapter 1 workbook for full claim table.

AI's draft output correctly:

- Refused to guess decision-impact for any item, marking each "NEEDS HUMAN" instead of inventing a weighting.
- Declined to fabricate an options-signal question since none had been supplied.
- Grounded the noise list directly in Pranavi's own CANNOT-KNOW.md exit triggers, rather than generic boilerplate.

Human decision-impact weighting supplied by Pranavi:

Claim	Question asked	Pranavi's answer	Interpretation
3a	If durable-goods spending growth does NOT extend to branded jewelry/watches, would you still hold the thesis?	No	Decision-changing — thesis's load-bearing claim, confirmed
2	If the real middle-class growth figure is meaningfully lower than 6.3%/yr, does that change your view?	Yes	Decision-changing — more important than initially assumed
4	Does the "jewelry as first upgrade" framing do real independent work, or is it just supporting color?	Maybe (personally yes, but electronics likely come first) — treated as No / low-impact	Deprioritized — the framing itself is suspect

Finalized effort-plan.md

Rank	Item	Decision-impact	Effort	Action
1	Claim 3a — durable-spending link to branded jewelry/watches	Yes — changes decision	High	Research first regardless of effort
2	Claim 2 — middle-class stat sourcing (secondhand via minister's speech)	Yes — changes decision	Low–Med	Research second — low effort, confirmed importance

Deprioritized (moved to BACKLOG.md): Claim 4 (jewelry as "classic first upgrade" category) — supporting color only, not independently load-bearing.

Do NOT spend time here (noise, per CANNOT-KNOW.md): daily gold-price moves; single-quarter volume dips (within the 2–3 quarter exit-trigger window); single-earnings-print reactions in isolation; 0DTE/short-dated options chatter (no options-signal question was ever named for this thesis).

Exercise 4 — CLI: effort-plan.md + BACKLOG.md

Run via Claude Code inside the chapter-1/chapter-2 project folder. Resulting structure:

```
chapter-1/  
■■■ theses/TITAN.md (read-only, untouched)  
■■■ CANNOT-KNOW.md (read-only, untouched)  
■■■ effort-plan.md (reformatted: decision-movers only)  
■■■ BACKLOG.md (new: deprioritized + noise, as checklist)
```

Inspection results:

Check	Result
Read-only files (TITAN.md, CANNOT-KNOW.md) untouched	Pass
No new claims, tickers, or judgments invented	Pass
No "verified" or "real signal" labels added by AI	Pass
Pranavi's own decision-impact wording preserved verbatim	Pass
Noise items relocated to BACKLOG.md, not deleted	Pass
Process stopped for review before further action	Pass

Exercise 5 — AI Validation Checklist

Checklist item	Result	Note
Correctness	Pass	Each "what would settle it" cites a real, openable source (next HCES round / GJEPC data; primary OECD/World Data Lab study)
Completeness	Pass	All open claims (3a, 2, 4) captured; none silently dropped
Scope	Pass	No recommendation, size, or price target added anywhere
Decision-impact	Pass	Top-ranked items reflect Pranavi's actual stated impact, not model-perceived salience; Claim 4 correctly demoted once real reasoning was supplied
Noise filter	Pass	Gold-price moves, single-quarter dips, single-earnings reactions, 0DTE chatter all correctly quarantined, grounded in CANNOT-KNOW.md
Failure-mode check	Pass	No fluent-but-wrong ranking; AI explicitly deferred decision-impact rather than guessing

Verdict: All six items pass. Per the exercise's rule, this plan is cleared to run the actual research session against — Claim 3a first, then Claim 2.

AI Use Disclosure

The AI produced an initial item inventory and draft ranking based on theses/TITAN.md and CANNOT-KNOW.md, correctly flagging every decision-impact cell as unresolved rather than guessing; Pranavi supplied her own weighting (yes/yes/maybe-leaning-no) to finalize the ranked order. The one thing the AI could not determine was which claims would actually flip the decision — that depends on personal risk tolerance and thesis weighting, which no model has access to.

Part 3 — Completion Status

Exercise Set	Status
LLM Exercises 1–3 (workflow labeling, uncertainty prompts, phase-gate design)	Complete
Your Own Mycroft — Exercise 1 (When to Use AI)	Complete
Your Own Mycroft — Exercise 2 (When NOT to Use AI)	Complete
Your Own Mycroft — Exercise 3 (Effort Plan draft & human-finalized)	Complete
Your Own Mycroft — Exercise 4 (CLI: effort-plan.md + BACKLOG.md)	Complete
Your Own Mycroft — Exercise 5 (AI Validation Checklist)	Complete

Chapter 2 — The Reallocation Principle — is fully complete.